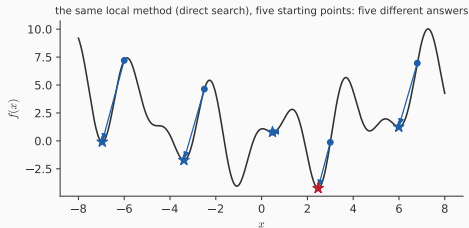


Optimization

5.1 GLODS — Global and Local Optimization using Direct Search

José Firmino Aguilar Madeira

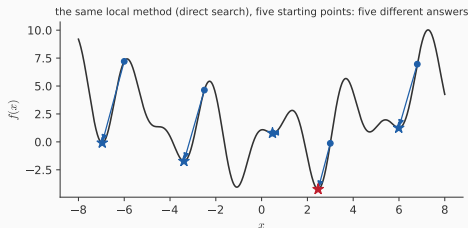
Chapter 5. Global optimization — local or global?



$f(x) = 0.1x^2 + 3 \sin 2x + 2 \cos(3.3x + 1)$ on $X = [-8, 8]$: the reference function for the whole chapter.

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Chapter 5. Global optimization — local or global?

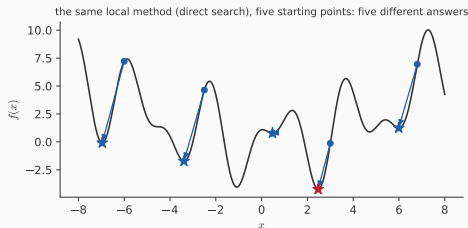


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Question: which of them solved the problem?

local minimum: $f(x^*) \leq f(x)$ for $x \in X$,
 $\|x - x^*\| < \delta$ (some $\delta > 0$)

global minimum: $f(x^*) \leq f(x)$ for all $x \in X$

Every global minimum is local; not conversely.

Converging does not mean finding the best solution of the problem.

Why search globally?

So far (Chs. 2–4) we have learned to **descend and converge**. In an engineering problem that is not enough:

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2. **We do not know how many basins there are:** with a black box we do not even see the landscape.
3. **One local solution may be much worse than another:** being stationary is not necessarily what the engineer wants.
4. **Several good minima are also information:** manufacturing, cost, robustness or maintenance may lead us to choose a solution that is not the one with the lowest f — we return to this at the end of 5.1.

But there is a price

If we do not know where the best basin is, we have to **explore**. And exploring **costs evaluations of f** .

Exploration

visiting new regions of X , looking for other basins

Refinement

descending within a basin already found, down to its minimum (exploitation)

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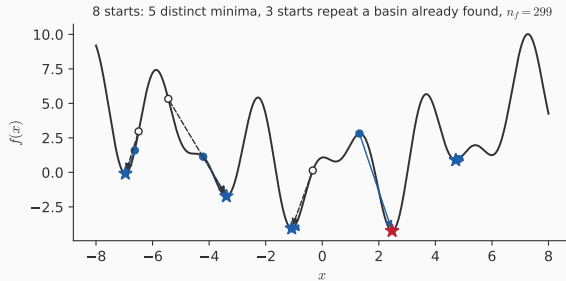
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Question: how can we explore different basins **without spending the whole budget**?

The goal of this chapter

Search globally: explore different basins and try to identify solutions better than that of a single local search. **Guaranteeing** the global optimum is another matter.

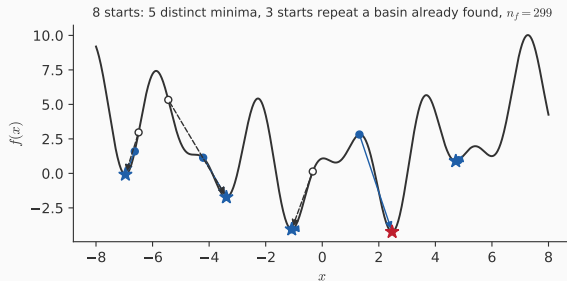
The simplest remedy: multistart



Multistart: run the same local method from several points and keep the best.

Here, 8 random starts found 5 distinct minima, including the global one.

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Question: what is the problem?

Three of the eight starts (dashed) fell into a basin already found and **repeated practically the same work**.

We need to explore the space without wasting too much budget.

Three ways to escape — the map of Chapter 5

How do we escape from a local basin — and how much does it cost us?

5.1 GLODS

Several **points** + local searches (direct search, 3.2.5) + avoiding redundancy. Deterministic (with a *deterministic search*); returns a *list* of minima.

5.2 Simulated annealing

One **point** that sometimes accepts **getting worse**. Stochastic.

5.3 Genetic algorithms

A **population**: selection + recombination + mutation. Stochastic.

At the end of each method, the same five questions:

Global exploration

how does it escape basins?

Information

what does it need from f ?

Cost

how many evaluations?

Guarantee

what can be claimed?

Randomness

is a run reproducible?

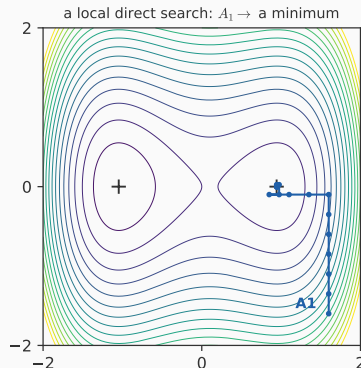
The common thread of the course — information, cost, guarantee — applied to the global question.

GLODS: multistart conditioned by proximity

GLODS

A multistart strategy **conditioned by proximity**: if two searches get close enough, **only one remains active**.

- A **direct search** (3.2.5) from A_1 tends to converge to the minimum of its basin.



Aluffi-Pentini on $[-2, 2]^2$ (the sequence from the 2015 presentation, redone).

Only values of f ; f expensive; X compact (extreme barrier).

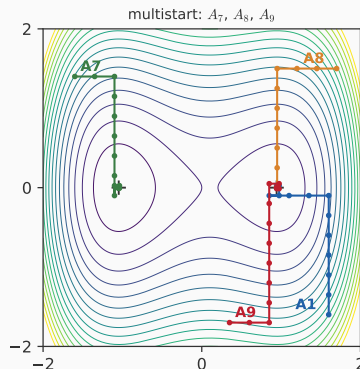
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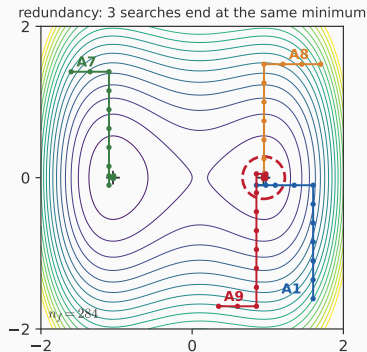
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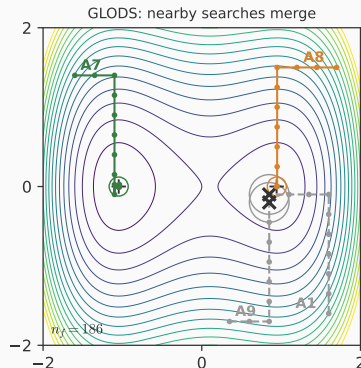
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- **Redundancy**: three end at the same minimum;
 $n_f = 284$ for 2 minima.
- **GLODS**: the same searches simultaneously; the one that enters the neighbourhood of a better one is no longer active ($A_9 \rightarrow A_1$, $A_1 \rightarrow A_8$). The same 2 minima, $n_f = 186$.



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The idea that sets GLODS apart: (x, α, r, i)

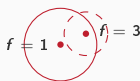
Each point in the list stores four things:

x position

α direct-search step size

r comparison radius

i active / inactive



far apart: two active searches close: one falls within the other's radius only the better one stays active

Rule (simple decrease)

A new point x_{new} is **comparable** with a stored point y when $\|x_{\text{new}} - y\| \leq r_y$. Comparable with none: new region, enters **active**. Better than all comparable points: enters **active** and deactivates the worse ones. Some comparable point as good or better: enters **inactive** (if it is better than some active one) or does not enter at all. An active point may become inactive; never the reverse. Only active points are *poll* centres.

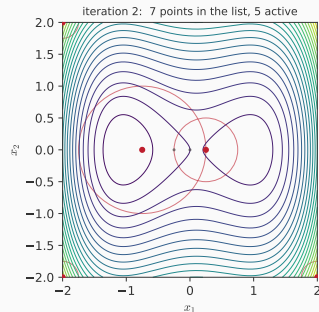
The radius comes from the **direct-search step size**, not from a probabilistic construction: $r = \text{step with which the point entered}$ ($\alpha \|d\|$; α with unit directions); it does not decrease when α decreases ($r \geq \alpha$).

When two searches get close: merging

The full GLODS, with a 2^n -centres *search*, on the same problem: the new points go through the comparison rule and the searches that get close to better ones are no longer active.

Three types of iteration

Successful: a new active point entered — the step is kept or increased (in the examples, $\times 2$). **Unsuccessful:** no point entered — the centre's step is reduced (in the examples, $\times \frac{1}{2}$). **Merging:** only inactive points entered — not a failure: proximity detected redundant work.



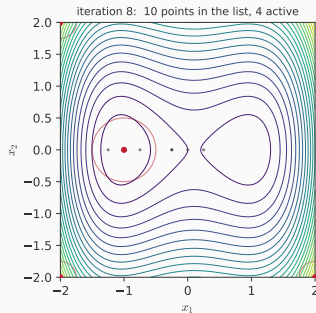
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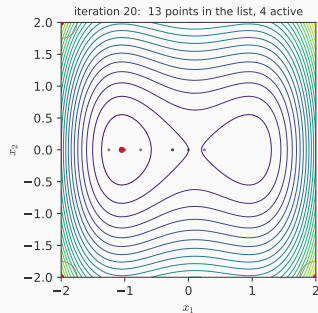
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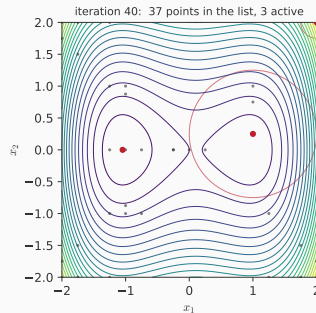
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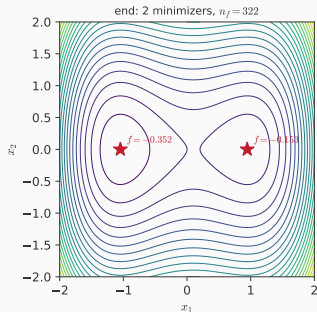
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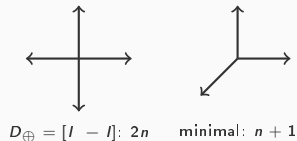


Aluffi-Pentini, $\alpha_0 = r_0 = 0.25$. Red: active, with their radius; grey: inactive. End: 2 minimizers, $n_f = 322$; 32 successful it., 29 unsuccessful, 10 merging. Multistart from 8 random points: in one run, the same 2 ($n_f = 848$); in 30 runs, $n_f = 850$ [823, 882] and both only in 12.

Poll: the local component (direct search, 3.2.5)

From x_k with step α_k , evaluate f at the **poll** points $x_k + \alpha_k d$, $d \in D_k$, where D_k is a **positive spanning set**: every vector of $\mathbb{R}^n$ is a *nonnegative* linear combination of elements of D_k . If $\nabla f(x_k) \neq 0$, some direction of D_k is a descent direction — without computing the gradient.

- Maximal positive basis $D_{\oplus} = [I \ -I]$: $2n$ directions ($\pm e_i$); minimal: $n + 1$ directions.
- **Success** (some poll point improves): the step is kept or grows, $\alpha_{k+1} \in [\alpha_k, \gamma\alpha_k]$; **failure**: it decreases, $\alpha_{k+1} \in [\beta_1\alpha_k, \beta_2\alpha_k]$, $0 < \beta_1 \leq \beta_2 < 1$. In the examples: $\times 2$ and $\times \frac{1}{2}$.
- Bounds $x^{(L)} \leq x \leq x^{(U)}$ by *extreme barrier*: $f = +\infty$ outside X (not evaluated; not to be confused with the logarithmic barrier of 4.3.2).
- Local stopping: $\alpha_k \leq \alpha_{\min}$. The poll step is the α_k of Chs. 2–3 and the P_j of Hooke–Jeeves; it returns in 6.5 (DMS).



Cost per iteration

$n_f \leq |D_k|$ ($2n$ with $D_{\oplus}$); only values of f . It is the “thermostat” of Hooke–Jeeves and Box; in the formalized versions (GPS, 3.2.5; here in GLODS) — not in the simplified Box of 3.2.4 — it gives stationarity results under suitable assumptions, which NM does not have.

Search: the global component

As in modern direct-search methods (patternsearch), the iteration is organized in two steps:

Search (optional, global)

Evaluate f at *any* finite set of points of the domain, looking for promising regions not yet explored: random sampling, Latin hypercube, Sobol/Halton sequences, or 2^n -centres (next page). **If it generates a new active point, the iteration is successful and ends without a poll.** It need not be done at every iteration: in the paper, the *search* is performed when the number of active points not yet identified as local minimizers is equal to 1.

Poll (local, structured)

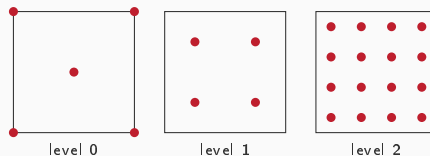
Explores an active region: the poll $x_k + \alpha_k d$, $d \in D_k$, around the active point chosen as centre (the one with the lowest f still to be explored). It is this step, with the structure of positive spanning sets and of the step size α_k , that **supports the convergence theory**; the *search* can be anything.

Every new point, whether from the *search* or the *poll*, goes through the same comparison rule (“The idea that sets GLODS apart”): it decides whether the point enters active, inactive, or not at all.

One option for the *search*: 2^n -centres

A **deterministic** strategy that meets three requirements: points asymptotically dense in X , uniform coverage and reproducibility.

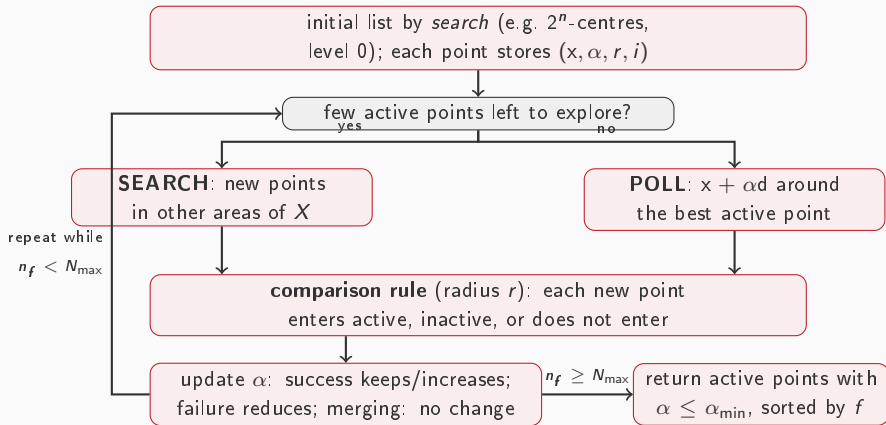
- **Level 0**: the centre *and the vertices* of the box ($2^n + 1$ points).
- **Level $\ell \geq 1$** : the box is subdivided into $(2^\ell)^n$ boxes; the points are their centres.
- To keep the cost from exploding, each *search* uses only a subset of 2^n centres not yet used; the rest are left for the following ones.
- Each new centre goes through the comparison rule (those that fall within the radius of a better point enter inactive or not at all). In the teaching version they enter with α_0, r_0 ; in the full algorithm, comparable points may inherit the parameters of the existing search.



$n = 2$: level 0, 5 points (centre + vertices); level 1, 4 centres; level 2, 16 centres, served 4 at a time.

Other search strategies: random sampling, Latin hypercube (LHS), Sobol and Halton sequences. GLODS is not the 2^n -centres: it is the *search/poll* framework with proximity; the *search* is a choice.

The full GLODS



The *poll* ensures convergence (theory); the *search* looks for new basins; the comparison rule avoids repeated work.

Algorithm (teaching version)

GLODS: $X, \alpha_0, r_0 \geq d_{\max}\alpha_0, \alpha_{\min}, N_{\max}$

```
1: list  $L \leftarrow$  initial search (level-0  $2^n$ -centres), each point with  $\alpha_0, r_0$ , after the
   comparison rule
2: while  $n_f < N_{\max}$  do
3:    $C \leftarrow \{x \in L \text{ active} : \alpha_x > \alpha_{\min}\}$   $\triangleright$  active points not yet identified as
   minima
4:   if  $|C| \leq 1$  then search:  $2^n$  new centres into  $L$ ; if one entered active,
   continue; otherwise recompute  $C$ 
5:   end if
6:   if  $C = \emptyset$  then continue (next search) or stop, if levels/budget are exhausted
7:   end if
8:    $x \leftarrow \arg \min_C f$   $\triangleright$  poll centre: the best one still to explore
9:   poll: for each  $d \in D$ , submit  $x + \alpha_x d$  to the comparison rule (step  $\alpha_x$ ,
   radius  $\alpha_x \|d\|$ )
10:  if some active point entered then successful:  $\alpha \leftarrow \gamma\alpha$  at the new active
   points
11:  else if some inactive point entered then merging: nothing changes
12:  else unsuccessful:  $\alpha_x \leftarrow \beta\alpha_x$ 
13:  end if
14: end while
15: return active points with  $\alpha \leq \alpha_{\min}$ , sorted by  $f$   $\triangleright$  local minimizers
```

- The output is a **list**, not a point: the first is the best minimum found. The cost is n_f ; the budget $N_{\max}$ is what the user controls.
- $r =$ step with which the point entered ($\alpha \|d\|$; α with unit directions); it does not decrease when α decreases ($r \geq \alpha$). In the paper, $r_0 \geq d_{\max}\alpha_0$.
- Deterministic with 2^n -centres; stochastic with random sampling (then: several runs, median).
- Teaching version: simple decrease, no greedy ordering of the list and no initial line sampling; the structure is that of the paper.

What can be guaranteed?

Theorem (refining subsequences) — Custódio and Madeira (2015)

Under the assumptions of the globalization strategy used (integer lattices with simple decrease, or sufficient decrease with a forcing function), X compact and f bounded below: $\liminf \alpha_k = 0$, and there is a convergent subsequence of unsuccessful *poll* iterates with $\alpha_k \rightarrow 0$. (Globalization in the sense of 3.3.1: from any x_0 , towards stationarity, not towards the global minimum.)

Theorem (stationarity)

If f is L -Lipschitz (R) near the limit x_* and the refining directions are dense in the tangent cone, x_* is **Clarke stationary**: $f^\circ(x_*; d) \geq 0$ for the directions of the tangent cone to X . If f is **strictly** differentiable at x_* : in the interior $\nabla f(x_*) = 0$; on the bounds, KKT. With a fixed $D_\oplus$ (as in the examples) only $f^\circ(x_*; d) \geq 0$ for $d \in D_\oplus$ is guaranteed; with strict differentiability this is enough. For f only Lipschitz, dense directions are needed.

$f^\circ(x; d) = \limsup_{y \rightarrow x, t \downarrow 0} [f(y + td) - f(y)]/t$: Clarke generalized directional derivative. Tangent cone to X at x : the directions along which one enters X from x . In the line of Torczon, Audet and Dennis, Kolda–Lewis–Torczon; full presentation.

What this says — and what it does not

- The theory provides **stationarity conditions for limit points of refining subsequences**; applying the method does not require smoothness (only the theorem requires local Lipschitz continuity).
- **Global exploration** comes from the *search*: how many basins are found depends on the budget, the sampling and the comparison radius. Without additional assumptions on f , there is no general guarantee of identifying exactly the global minimum with a finite budget.
- Noise: GLODS does not need derivatives, which noise would make unreliable; but this is *not* a theory for noisy functions.

What the 2015 paper showed

61 bound-constrained problems, $2 \leq n \leq 10$, budget 20 000, stopping at $\alpha < 10^{-8}$; data profiles against **DIRECT**, **MCS** and a pure multistart. The conclusion is balanced:

- pure multistart: poor performance (the waste seen at the start of the deck); DIRECT behind MCS and GLODS;
- *small* budgets: MCS was the best on the test set;
- GLODS **competitive**, with the best performance for *moderate* budgets; with random initialization, the advantage of including a *search* became clear — 2^n -centres, Halton and Sobol performed well;
- and the additional feature: looking for **the different local minima**, not just the best value.

Problem (Tab. 6)	n	known min.	local	found
Becker-Lago	2	4		4
Branin-Hoo	2	3		2–3
Aluffi-Pentini	2	2		1
Shekel-45	4	5		4–5
Six-hump camel	2	6		2
Periodic	2	50		6–8

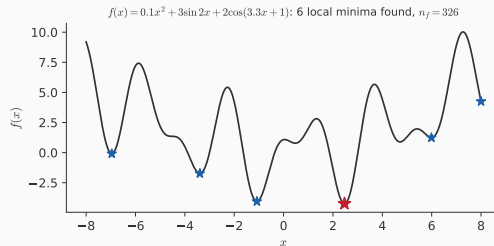
Variants 2^n -centres, Sobol, Halton, with $\alpha_0 = r_0 = 1$ on the original domains. **GLODS tries to explore different basins; budget, search and comparison radius determine how many it can identify.** On Aluffi-Pentini the paper found 1 of the 2; in the teaching example, varying r_0 ($r_0 = 1$ gives 1, $r_0 = 0.25$ gives 2) shows how the radius can influence the separation of basins.

GLODS on the Chapter 5 function

$f(x) = 0.1x^2 + 3\sin 2x + 2\cos(3.3x + 1)$ on $[-8, 8]$;
 2^n -centres up to level 4, $\alpha_0 = r_0 = 1$, $\alpha_{\min} = 10^{-4}$,
 $n_f = 326$:

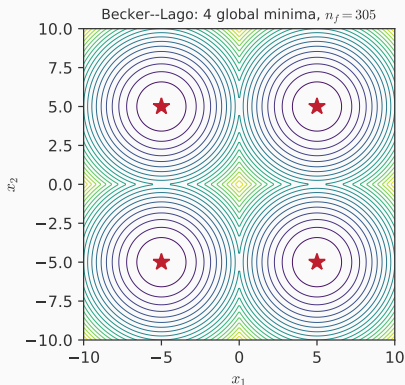
	x	$f(x)$
1	2.4680	-4.2381
2	-1.0819	-4.0531
3	-3.3918	-1.7273
4	-6.9615	-0.0849
5	5.9908	1.2485
6	8.0000	4.2533

- The minima at 0.47 and 4.72 were **absorbed**: points from those basins fell within the comparison radius of neighbours with lower f . Depth alone does not decide (4.72 is lower than 5.99): order and radius also count.



- Five of the seven interior local minima, with the global one $x^* = 2.468$, $f^* = -4.238$ **first**.
- $x = 8$: boundary minimum (f decreases up to the bound; stationary in the Clarke-KKT sense).

Several global minima: Becker–Lago



$$f = (|x_1| - 5)^2 + (|x_2| - 5)^2 \text{ on } [-10, 10]^2.$$

- Four global minima, all with $f = 0$, at $(\pm 5, \pm 5)$.
- GLODS finds all four with $n_f = 305$ ($\alpha_0 = r_0 = 2$) — as in the paper (4 of 4).
- f is nondifferentiable on the axes: the algorithm applies without derivatives; Clarke analysis is the appropriate framework for interpreting stationarity.
- A method that returns a single point would give *one* of the four — and we would not know there are three others.

On the computer: the GLODS MATLAB code

The original code (version 0.2) is distributed in class. It is not meant to be read from end to end: what matters is recognizing the pieces of the method in it.

Idea of the method	File
<i>search</i> (global component)	<code>search_step.m</code>
proximity: comparison radius and list of active/inactive points	<code>match_point.m</code> , <code>merge.m</code>
main algorithm (<i>search/poll</i> , update of α)	<code>glods.m</code>
options (<i>search</i> strategy, α_0 , r_0 , stopping, budget)	<code>parameters_glods.m</code>
starting point : a simple call on a test problem	<code>driver_glods.m</code>

- Start with `driver_glods.m`: run it, change the *search* strategy and the initial radius, and see how many minima appear in the list.
- Only then open `glods.m`, with the flowchart “The full GLODS” alongside.

Optimization of the fibre orientation in a laminated composite specimen (Monte, Infante, Madeira, Moleiro, 2017)

Decision the fibre orientation in each of the 8 plies: angles in $[-85^\circ, 90^\circ]$, in increments of 5° (36 values per ply).

Objective minimize the maximum displacement of the specimen under load.

Information on f values only: each evaluation is a finite element analysis (ANSYS), called by GLODS — black box, expensive, no derivatives.

Result a list of 14 solutions with maximum displacements between 3.17 and 3.60 mm — all of them solve the problem.

The final decision was not “the one with the lowest f ”: among solutions with equivalent performance, **the easiest to manufacture (lay up) was chosen**. A method that returned only one solution would not naturally provide this set of alternatives in a single run. Reference: *Mechanics of Advanced Materials and Structures* 24(5):410–416, 2017, DOI 10.1080/15376494.2016.1191099.

Is the global minimum always the chosen solution?

Two of the 14 composite solutions:

Solution 12 — the best f

$[-35/45/40/80/ - 40/50/ - 55/40]$

maximum displacement **3.17 mm**

eight different orientations

Solution 1

$[45/ - 50/ - 50/45/45/ - 50/ - 50/45]$

maximum displacement **3.40 mm** (+7%)

only two orientations, in a symmetric pattern

Is the global minimum always the chosen solution?

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only two orientations, in a symmetric pattern

Question: which one would you rather lay up?

Is the global minimum always the chosen solution?

Two of the 14 composite solutions:

Solution 12 — the best f

$[-35/45/40/80/ - 40/50/ - 55/40]$

maximum displacement **3.17 mm**

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Solution 1

$[45/ - 50/ - 50/45/45/ - 50/ - 50/45]$

maximum displacement **3.40 mm** (+7%)

only two orientations, in a symmetric pattern

Question: which one would you rather lay up?

The mathematical optimum is the **best according to the model** we formulated.

The engineering decision may use **information that did not enter the model**.

If we choose another solution for cost, manufacturing or materials, this reveals that there were relevant criteria outside f . If the algorithm had returned only solution 12, we would not even know that solution 1 existed.

So why find several minima?

Because different basins may correspond to **qualitatively different design alternatives**.

- **Similar performance** → nearly equivalent alternatives.
- **Manufacturing, cost, materials** → criteria that are not in f .
- **Robustness of the solution** (not of the method) → a wide basin (small changes in x barely degrade f) may be preferable to a slightly better narrow minimum.
- **Later decision** → the engineer chooses among alternatives, list in hand.

An algorithm that returns only the best point answers “what is the best solution according to f ?”. A list also answers “**what alternatives do I have?**”.

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And if those criteria are important enough?

Perhaps they should not stay outside the model: they become **objectives**.

performance → (performance, cost,
manufacturability, ...)

⇒ **Ch. 6: several objectives**.

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And with several objectives?

MultiGLODS extends this global–local idea to multiobjective optimization (Custódio and Madeira, *J. Global Optim.* 72, 2018). We do not study it in this course; we return to several objectives in Ch. 6.

5.1 GLODS — summary table

What information do we have about f ? What does it cost to obtain? What can we guarantee?

Information	only values of f (order 0); bounds $x^{(L)} \leq x \leq x^{(U)}$ by extreme barrier
Global exploration	several local direct searches (<i>poll</i>) + <i>search</i> in unexplored areas, without duplicating nearby searches: comparison radius and list of active/inactive points
Cost	evaluations of f : $n_f \leq D $ per iteration ($2n$ with $[I - I]$) plus those of the <i>search</i> ; the budget $N_{\max}$ is the user's decision
Guarantee	(Clarke) stationarity of limit points of refining subsequences, if f is Lipschitz near the limit and the refining directions are dense (with a fixed $D_{\oplus}$: only along the directions of $D_{\oplus}$); no guarantee of finding all basins with finite n_f
Output	a <i>list</i> of local-minimizer candidates, sorted by f ; allows deciding by criteria that were not in f
Randomness	can be deterministic (2^n -centres; deterministic sequences) or stochastic (random sampling, LHS); in the latter case it depends on the seed and requires several runs
Limitation	high dimension ($2n$ poll points per iteration, 2^n -centres) and very expensive functions require managing the budget; small basins may be absorbed by the radius

KEY TAKEAWAYS — 5.1

1. **Global exploration:** local *poll* + *search* in new areas; the comparison radius avoids repeating searches.
2. **Information and cost:** only values of f ; cost in evaluations of f , with a fixed budget.
3. **Guarantee:** (Clarke) stationarity of refining limits, if f is Lipschitz near the limit and the refining directions are dense (with a fixed $D_{\oplus}$: only along the directions of $D_{\oplus}$); no guarantee of finding all basins.
4. **Output:** a **list** of local-minimizer candidates — so that we can also choose by criteria outside f .
5. **Randomness:** deterministic with a deterministic *search*; otherwise, several runs.

Next (5.2, 5.3): randomness — SA accepting uphill moves, the GA with a population.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Custódio, A. L., Madeira, J. F. A., *GLODS: Global and Local Optimization using Direct Search*, Journal of Global Optimization 62(1):1–28, 2015 — the GLODS paper.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 7.
- Madeira, J. F. A., *Multiobjective and global optimization with directional direct search*, PhD thesis in Mathematics, IST, Universidade de Lisboa, 2019.
- Horst, R., Tuy, H., *Global Optimization: Deterministic Approaches*, 3rd ed., Springer, 1996.

Resources

- Extensions: Madeira (2026) (GLODS-SI, scale invariance); *GLODS-SI-Mix* — continuous, discrete and categorical variables simultaneously (submitted).
- The **full presentation** (98 pages): convergence with Clarke–Jahn derivatives, data profiles against DIRECT and MCS. These lecture notes are the teaching version.
- GLODS v0.2 (MATLAB), distributed in class with the technical report (files in “On the computer”); page: docentes.fct.unl.pt/algb/pages/glods.